

End-term Exam 2025, BDA

All questions are compulsory. Show all the steps of your solution.

Time: 2 hrs

Total marks: 30

Roll Number:

Name:

Q1: [4 marks] Given data files of two relations Employee(eid, name, salary, pid) and Projects(pid, pname, deptid), write the pseudo code for mappers and reducers for the following queries. Employee relation has the information of employees along with their project pids that employee has worked on. Projects relation has the information of projects along with the controlling department deptid. Primary keys of the relations are underlined.

- A) Find name of employees who have worked on projects controlled by a department with deptid='x'. [2 Marks]

map_EP(key, line):

```
if (key == 'Employee'):
    // line format: eid, name, salary, pid
    (eid, name, salary, pid) = parseEmployee(line)
    emit(pid, ("E", name))
else:
    // line format: pid, pname, deptid
    (pid, pname, deptid) = parseProject(line)
    if deptid == 'x':
        emit(pid, ("P"))
```

Reduce(pid, values):

```
hasDeptXProject = false
empNames = [ ]
// values is a list of tagged tuples:
// ("E", name) from Employee
// ("P")    from Projects (only deptid='x' filtered in mapper)
for v in values:
    if v[1] == "P":
        hasDeptXProject = true
    elif v.tag == "E":
        empNames.append(v.name)
if hasDeptXProject:
    // Output all employees who worked on this pid
    for name in empNames:
        emit(name, null)
```

- B) Find departments who controlled more than 10 projects that are associated with more than 5 employees. [2 Marks]

This would require two MR jobs. Job-1

Map1(key, line):

```
if (key == 'Employee'):
    // line format: eid, name, salary, pid
    (eid, name, salary, pid) = parseEmployee(line)
    emit(pid, ("E", eid))
else:
    // line format: pid, pname, deptid
    (pid, pname, deptid) = parseProject(line)
    emit(pid, ("P", deptid))
```

Reducer1(key, value):

```
deptid = null
empSet = [ ]    // to ensure distinct employees

for v in values:
    if v[0] == "P":
        deptid = v.deptid
    else:
        empSet.add(v.eid)

if deptid != null and size(empSet) > 5:
    // This project (pid) qualifies; count 1 for its department
    emit(deptid, 1)
```

Job- 2:

Map2(deptid, one):

```
emit(deptid, 1)
```

Reduce2(deptid, values):

```
projectCount = len(values)
if projectCount > 10:
    emit(deptid, projectCount)
```

Q2: [4 marks] Consider the following matrix:

Row, C1, C2, C3, C4

R1, 0, 1, 1, 0
R2, 1, 0, 1, 1
R3, 0, 1, 0, 1
R4, 0, 0, 1, 0
R5, 1, 0, 1, 0
R6, 0, 1, 0, 0

Perform a minhashing of the data, with the order of rows taken as R4, R6, R1, R3, R5, R2. The row R4 becomes the first row, R6 becomes the second row and so on. **Note:** we give the minhash value in terms of the original name of the row, rather than the order of the row in the permutation. For an example if the output is 3rd row, you shall output row label R1.

- a) Give the min-hash value for each of the columns C1, C2, and C3. [3 marks]

C1 → **R5**

C2 → **R6**

C3 → **R4**

- b) Give an ordering/permutation of rows such that C1 and C3 get different min-hash values [1 marks]. In case it is not possible, please write NOT POSSIBLE.

R1, R2, R3, R4, R5, R6

Any permutation which has R1 and R4 before R2 and R5 would result into different minhash values.

Q3: [4 marks] Suppose we have computed signatures for a number of columns, and each signature consists of 24 integers, arranged as a column of 24 rows. There are N pairs of signatures that are 50% similar (i.e., they agree in half of the rows). There are M pairs that are 10% similar, and all other pairs (an unknown number) are 0% similar. We can try to find 50%-similar pairs by using Locality-Sensitive Hashing (LSH), and we can do so by choosing bands of 1, 2, 3, 4, 6, 8, 12, or 24 rows. Then, suppose that we assign equal cost to false positives and false negatives (an atypical assumption).

- A) Give an expression for finding the band size if $M=100N$? [4 marks]

For a pair with similarity s , the probability that it becomes a candidate under LSH is

$$P_{\text{cand}}(s) = 1 - (1 - s^r)^b$$

For **50%-similar pairs** ($s = 0.5$):

- Detection (candidate) prob: $P_{\text{det}} = 1 - (1 - 0.5^r)^b$
- False negative prob: $P_{\text{FN}} = (1 - 0.5^r)^b$

For **10%-similar pairs** ($s = 0.1$): False positive prob: $P_{\text{FP}} = 1 - (1 - 0.1^r)^b$

Total expected error (false positives + false negatives), with equal cost per error, is

$$E(r) = N \cdot P_{\text{FN}} + M \cdot P_{\text{FP}} = N(1 - 0.5^r)^b + M[1 - (1 - 0.1^r)^b]$$

Given $M = 100N$ and $b = 24/r$, this becomes

$$E(r) = N[(1 - 0.5^r)^{24/r} + 100(1 - (1 - 0.1^r)^{24/r})].$$

Final Expression:

Choose the band size $r \in \{1, 2, 3, 4, 6, 8, 12, 24\}$ that **minimizes**

$$(1 - 0.5^r)^{24/r} + 100(1 - (1 - 0.1^r)^{24/r}).$$

Q4: [4 marks] Suppose we have an LSH family h of $(d1, d2, .6, .4)$ hash functions. We use two functions from h and create the AND-construction to form a $(d1, d2, w, x)$ family. Similarly, we use three functions from h and create the OR-construction to form a $(d1, d2, y, z)$ family. What are the values of w, x, y and z ? [1+1+1+1 marks]

Ans:

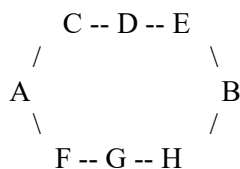
$$\begin{aligned} w &= 0.36 \\ x &= 0.16 \\ y &= 0.936 \\ z &= 0.784 \end{aligned}$$

Q5: [3 marks] Suppose we are using the DGIM algorithm to estimate the number of 1's in suffixes of a sliding window of length 45. The current timestamp is 100, and we have the following buckets stored in format (timestamp, size): (100, 1), (98, 1), (95, 2), (92, 2), (87, 4), (80, 4), (60, 8) Note: Shown timestamps are absolute values, rather than modulo the window size, as DGIM would do. Suppose that at times 101 through 105, 1's appear in the stream. Compute the set of buckets that would exist in the system at time 105.

Answer:

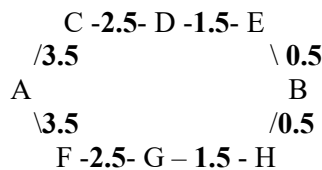
Buckets at 101: (101,1), (100,2), (95,4), (87,8), (60,8)
 Buckets at 102: (102,1), (101,1), (100,2), (95,4), (87,8), (60,8)
 Buckets at 103: (103,1), (102,2), (100,2), (95,4), (87,8), (60,8)
 Buckets at 104: (104,1), (103,1), (102,2), (100,2), (95,4), (87,8), (60,8)
 Final answer: buckets at time 105: (105,1), (104,2), (102,4), (95,4), (87,8)

Q6: [2 marks] For the following graph:



Apply the Girvan-Newman algorithm with root as node A. Compute the credit for each edge. What is the credit of edges BH and AF? [1 +1 marks]

Answer:



$$\mathbf{BH = 0.5}$$

$$\mathbf{AF = 3.5}$$

Q7: [3 marks] Suppose the Web consists of a clique (set of nodes with all possible arcs from one to another) of n nodes and a single additional node that is the successor of each of the n nodes in the clique. We want to determine the PageRank of each page, as a function of n and β . While computing Pagerank, assume the Pagerank of each of the node in the clique as x and the Pagerank for the extra added node as y .

a) Give equations for x and y . [2 marks]

$$x = \beta * \left(\frac{(n-1) * x}{n} + \frac{y}{n+1} \right) + \frac{1-\beta}{n+1}$$

$$y = \beta x + \frac{1-\beta}{n+1}$$

b) What are specific values of x and y when β is 0.75? [1 marks]

$$x = \frac{n}{n^2 + n + 0.75}$$

$$y = \frac{(n+0.75)}{n^2 + n + 0.75}$$

Q8. [2 marks] Here is an example of a tiny Bloom filter. It uses an array of 10 bits and two independent hash functions f and g . We want to test membership in a set S of five elements, so we hash each of the five elements using both f and g , and we set to 1 any bit that any of the five elements is hashed to by either of the hash functions. When a new element x arrives, we compute $f(x)$ and $g(x)$, and we say x is in the set S if both $f(x)$ and $g(x)$ are 1. Assume x is not in the set S .

A) Give an expression for computing the probability of false positive, the probability of saying that x is in S when it is not there. [1 marks]

$$\mathbf{FP = \left(1 - \left(1 - \frac{1}{10} \right)^{10} \right)^2}$$

B) What is the false positive rate value? [1 marks]

$$\mathbf{1 - e^{-1} = 1 - 0.3679 = 0.6321.}$$

$$\mathbf{FP \approx (0.6321)^2 = 0.3996.}$$

Q9: [2 marks] We have the following user-item rating matrix (blank = missing):

U1: I1 = 5, I2 = 3, I3 = –

U2: I1 = 4, I2 = –, I3 = 2

U3: I1 = –, I2 = 1, I3 = 2

We use a matrix-factorization model with biases:

$$r_{i,u} = \mu + b_u + b_i + p_u^T q_i$$

where

- μ is the global mean rating
- b_u is the user bias
- b_i is the item bias
- p_u, q_i are 2-dimensional latent vectors

Latent vectors are given as:

$$p_{U1} = (1, 2), p_{U2} = (0, 1), p_{U3} = (2, 1)$$

$$q_{I1} = (1, 0), q_{I2} = (0, 1), q_{I3} = (1, 1)$$

Biases are estimated by simple mean deviation. What is the missing rating for user U_2 on item I_2 ? [2 marks]

Ans.

$$\text{Global mean } \mu = 17 / 6 \approx 2.83$$

$$\text{User biases: } b_{U1} = 4 - 2.83 = 1.17 \quad b_{U2} = 3 - 2.83 = 0.17 \quad b_{U3} = 1.5 - 2.83 = -1.33$$

$$\text{Item biases: } b_{I1} = 4.5 - 2.83 = 1.67 \quad b_{I2} = 2 - 2.83 = -0.83 \quad b_{I3} = 2 - 2.83 = -0.83$$

$$p_{U2} \cdot q_{I2} = 0 \cdot 0 + 1 \cdot 1 = 1$$

$$r_{I2}^{U2} = 2.83 + 0.17 - 0.83 + 1 = 3.17$$